WE Credit: Optimize Process Water Use

This credit applies to

- BD+C: New Construction (1-2 points)
- BD+C: Core & Shell (1-3 points)
- BD+C: Schools (1-2 points)
- BD+C: Retail (1-2 points)
- BD+C: Data Centers (1-2 points)
- BD+C: Warehouses & Distribution Centers (1-2 points)
- BD+C: Hospitality (1-2 points)
- BD+C: Healthcare (1-2 points)

INTENT

To conserve low cost potable water resources used for mechanical processes while controlling, corrosion and scale in the condenser water system.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE,

Option 1. Cooling Tower and Evaporative Condenser Cycles of Concentration (1-2 points except CS, 1-3 points CS)

For cooling towers and evaporative condensers, conduct a one-time potable water analysis, measuring at least the five control parameters listed in Table 1.

Table 1. Maximum concentrations for parameters in condenser water

Parameter	*Maximum level*
Ca (as $CaCO_3$)	600 ppm
Total alkalinity	500 ppm
SiO_2	150 ppm
Cl^-	300 ppm
Conductivity	3300 µS/cm

ppm = parts per million
µS/cm = micro siemens per centimeter

Calculate the maximum number of cooling tower cycles by dividing the maximum allowed concentration level of each parameter by the actual concentration level of each parameter found in the potable makeup water analysis. Limit cooling tower cycles to avoid exceeding maximum values for any of these parameters.

The materials of construction for the water system that come in contact with the cooling tower water shall be of the type that can operate and be maintained within the limits established in Table 1.

Table 2. Points for cooling tower cycles

Cooling tower cycles	*Points (all except CS)*	*Points (CS)*

Maximum number of cycles achieved without exceeding any maximum concentration levels or affecting operation of condenser water system	1	1
Meet the maximum calculated number of cycles to earn 1 point, and increase the number of cycles by a minimum of 25% by increasing the level of treatment and/or maintenance in condenser or make-up water systems OR Meet the maximum calculated number of cycles to earn 1 point and use a minimum 20% recycled nonpotable water	2	2
Meet the maximum calculated number of cycles to earn 1 point, and increase the number of cycles by a minimum of 30% by increasing the level of treatment and/or maintenance in condenser or make-up water systems OR Meet the maximum calculated number of cycles to earn 1 point and use a minimum 30% recycled nonpotable water	--	3

Minimum percentage recycled nonpotable water used in cooling tower makeup should be based on water use during the month with the highest demand for make-up water.

Projects may consider using water reclaimed from another process as a source of makeup water for evaporative cooling equipment as long as the resultant circulating water chemistry conforms to the parameters established in Table 1.

Projects whose cooling is provided by district cooling systems are eligible to achieve Option 1 if the district cooling system complies with the above requirements.

OR

Option 2. Optimize Water Use for Cooling (1-2 points except CS, 1-3 points CS)

To be eligible for Option 2, the baseline system designated for the building using ASHRAE 90.1-2016 Appendix G Table G3.1.1 must include a cooling tower (systems 7, 8, 11, 12, and 13)

Achieve increasing levels of cooling tower water efficiency beyond a water-cooled chiller system with axial variable-speed fan cooling towers having a maximum drift of 0.002% of recirculated water volume and three cooling tower cycles. Points are awarded according to Table 3.

Table 3. Points for reducing annual water use compared to Water-Cooled Chiller System

Percentage Reduction	Points (BD+C)	Points (CS)
25%	1	1
50%	2	2
100%	-	3

Projects whose cooling is provided by district cooling systems are eligible to achieve Option 2 if the district cooling system complies with the above requirements.

AND/OR

Option 3. Process Water Use (1-2 points except CS, 1-3 points CS)

Demonstrate that the project is using minimum 20% recycled alternative water to meet process water demand for 1 point, or using minimum 30% recycled alternative water to meet process water demand for 2 points. Ensure that recycled alternative water is of sufficient quality for its intended end use.

Minimum percentage of recycled alternative water used should be based on water use during the month with the highest water demand.

Process water uses eligible for achievement of Option 3 must represent at least 10% of total building regulated water use and may not include water used for cooling. Eligible subsystems may include:

- Boilers
- Humidification systems
- Other subsystems using process water

Projects served by district systems are eligible to achieve Option 3 if the district system complies with minimum thresholds for recycled alternative water use.

Core and Shell projects:
Demonstrate that the project is using minimum 20% recycled alternative water to meet process water demand for 1 point, using minimum 30% recycled alternative water to meet process water demand for 2 points, or using minimum 40% recycled alternative water to meet process water demand for 3 points. Ensure that recycled alternative water is of sufficient quality for its intended end use.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

Evaporative cooling is the most energy efficient method to reject heat from a process (1 lb. of water removes almost 1,000 BTU's (970 BTU/lb.) with the added benefit of reducing greenhouse gas emissions. Cooling towers and evaporative condensers utilize evaporative cooling to operate with maximum energy efficiency; to preserve this energy savings, it is important to conserve the water these system use, especially in water stressed regions. In some climate zones, heat rejection system designs can help to decrease the water intensity of the building cooling process using a combination of wet, dry, or adiabatic cooling. In some cases, 100% dry operation is possible.

For wet cooling systems, this credit addresses water conservation strategies while still saving energy. This strategy includes maximizing cycles of concentration in cooling towers and evaporative condensers and make-up water reduction strategies including the use of reclaimed water and hybrid cooling.
The use of alternative recycled water to meet process water demands can further reduce building potable water use.

Updates to this credit recognize regional variations in the quality of local water supply and recognize additional strategies for reducing potable water use for diverse project types.

Core and Shell projects can now earn up to 3 points under WE credit Optimize Process Water Use. One point was re-allocated from WE credit Indoor Water Use Reduction to this credit, to better align WE points available with the typical scope of work of Core and Shell projects and to reward incremental process water savings.

Step-by-Step Guidance

Option 1. Cooling Tower and Evaporative Condenser Cycles of Concentration

Refer to the v4 BD+C reference guide with the following addition:

Step 6. Address construction materials for cooling system

Share the results of the potable water analysis with the cooling system manufacturer and ensure that the materials of construction for all components of the water-cooled system that come in contact with the cooling loop can operate and be maintained within the limits established in Table 1. The evaporative cooled system equipment must be designed to accommodate the targeted cycles of concentration calculated for the cooling tower or evaporative condenser.

Option 2. Optimize Water Use for Cooling

Option 2 is limited to projects that use water-cooled systems in the ASHRAE 90.1-2016 baseline design. ASHRAE 90.1 Appendix G identifies baseline system types that are standard practice for similar, newly constructed buildings.

Step 1. Determine baseline system for building cooling.

Confirm that at least one of the baseline system(s) designated for the building using ASHRAE 90.1-2016 Appendix G Table G3.1.1 includes a cooling tower or evaporative condenser (systems 7, 8, 11, 12, and 13).

If the above conditions are met and the project team eliminates the need for cooling towers or evaporative condensers in the proposed building design, then the project team may document a 100% reduction in water use for cooling.

Step 2. Develop building cooling systems design.

Consider viable on-site alternatives to 100% evaporative cooled water-cooled systems for the building and the impacts of the cooling systems design on building energy use. Hybrid water-cooled system options may be used to reduce water usage while saving energy.

Step 3. Document compliance.

If the project includes systems that use the latent heat of evaporative cooling, calculate annual water savings as compared to a 100% water-cooled chiller system, based on proposed cooling loads for spaces served by systems 7, 8, 11, 12, or 13 in the baseline. The system performance for the reference shall be calculated based on a 100% water-cooled chiller system with axial variable-speed fan cooling towers having a maximum drift of 0.002% of recirculated water volume and three cooling tower cycles.

Some energy simulation software has the capability of simulating cooling tower water usage within the software. To perform the calculations within the energy model, the proposed case model shall be modeled consistent with the project design to demonstrate the total water usage associated with evaporative cooling in the proposed design.

To determine the reference case performance, the proposed case model shall be updated with a water-cooled chilled water system serving all spaces served by systems 7, 8, 11, 12, or 13 in the ASHRAE 90.1 Baseline model. All proposed parameters (such as fan power, fan control, economizer control, demand control ventilation, energy recovery, dedicated outside air supply, quantity of systems, etc.) shall remain identical, except that the cooling type shall be revised to water-cooled chillers matching those stipulated in ASHRAE 90.1 Appendix G.

If performing the cooling tower water usage calculations outside the energy model, cooling tower and evaporative condenser manufacturers have water usage programs based on design conditions, building load and water chemistry. To calculate the water usage with the manufacturer's data, the design cooling loads from the proposed model for spaces served by systems 7, 8, 11, 12, or 13 in the baseline shall be exported from the energy model, and supplemental calculations shall be documented based on the proposed building cooling load and water use data sheets from equipment suppliers. Any assumptions used for the calculations shall be submitted with the calculations.

The manufacturer's program used for calculating water usage must incorporate the hourly load profile from the energy model or be based on climate-specific load profiles for the project's building type. If hourly load profiles and hourly weather profiles are not used, the calculations shall include "bins" that account for variations in the load profile and water usage dependent on outdoor air drybulb temperature, outdoor air wetbulb temperature, and building cooling load. The load profile must be identical in the proposed and reference case.

Option 3. Process Water Use

Step 1. Identify water subsystems.
Eligible subsystems may include boilers, humidification systems, or other subsystems using process water that represent at least 10% of total regulated building water use. Water used for cooling is not eligible for Option 3.

- Consider minimum required quality of make-up water for water subsystems.

Step 2. Identify sources of recycled alternative water.
Identify recycled alternative water sources that could be used to meet process water subsystem demand.

Alternatives water sources include municipally supplied reclaimed wastewater ("purple pipe" water), graywater, rainwater, stormwater, treated seawater, water recovered from condensate, foundation dewatering water, treated blowdown from process water, reverse osmosis reject water, and other recycled water sources. Well water, groundwater, and naturally occurring surface bodies of water (such as streams, lakes, or rivers) do not contribute to recycled alternative water sources.

Step 3. Calculate subsystem water use.
Calculate the water use of subsystems using manufacturer information and anticipated operating conditions.

- To be eligible for Option 3, the subsystem must represent at least 10% of total building regulated water use annually. Total building water includes all indoor water consumption associated with the project in the design case and excludes outdoor landscape water use.
- Calculate subsystem water use during each month; identify the month with the highest water demand.

Step 4. Calculate quantity of recycled alternative water source.
Calculate the quantity of alternative water sources available for reuse each month. Address the cistern storage capacity for on-site alternative water systems. For municipally supplied alternative water systems, demonstrate that the municipality has approved to supply the volume of recycled water required by the project.

Step 5. Select alternative water source to meet subsystem demand.
Confirm that alternative water source is of sufficient quantity to meet at least 20% of water subsystem demand during the month with the highest water demand. If the alternative water is used for multiple applications—for example, boilers, flush fixtures, and landscape irrigation—a sufficient quantity must be available to meet the demands of all uses. Teams cannot apply the same alternative water to multiple credits unless the recycled alternative water source has sufficient volume to cover the demand of all the uses (e.g., boilers, irrigation plus toilet-flushing demand).

Confirm that alternative water source is of sufficient quality to meet intended use or treat alternative water source.

- Minimum requirements for make-up water quality vary by subsystem.
- When selecting alternative sources of water, target the uses that require the least treatment first.
- As needed, treat alternative water sources to be of sufficient quality for intended end use.

Project Type Variations

District Energy Systems

Option 2. Optimize Water Use for Cooling

For projects connected to a district cooling system, the project shall demonstrate that any on-site cooling equipment meets the credit requirements and that upstream systems meet the credit requirements.

To pursue this option, the baseline system designated for the building in accordance with ASHRAE 90.1-2016 Appendix G Table G3.1.1 (prior to modeling purchased energy) must include a cooling tower (systems 7 &, 8, 11, 12, and 13).

For upstream equipment from the District Energy System, provide a narrative describing how the system is operated and specifically describing the district energy cooling equipment, including capacities, equipment type, etc.

If cooling towers, evaporative condensers, evaporative cooling systems, and other evaporative cooling systems are not used in the district cooling system serving the project, or are used in conjunction with compressor-based cooling equipment to supply no more than 10% of the total cooling energy generated by the District Energy System, then the project automatically qualifies for 2 points under the credit (Core and Shell projects automatically qualify for 3 points) and does not need to provide calculations. Provide a narrative description of how the district cooling system is operated and confirm that there is no evaporative cooling equipment in the district energy system.

If the district cooling system does use evaporative cooling, the project team must provide calculations demonstrating achievement of the Option 2 requirements based on total district cooling capacity and cooling load profile for the entire district system (building cooling loads shall not be used to document performance).

The program or supplemental calculations used for calculating water usage must either incorporate the hourly load profile from the District Energy System, or include "bins" that account for variations in the District Energy Plant load profile and water usage dependent on outdoor air drybulb temperature, outdoor air wetbulb temperature, and District Energy thermal cooling load. The load profile must be identical in the proposed and reference case.

Option 3. Process Water Use

For projects where heating is provided by a district heating system, the project team must pro-rate the district heating system's total water usage to the building to determine whether it represents at least 10% of total building regulated water use.

For example, if the district heating system consumes 500,000 gallons of water annually and the LEED project uses 1% of the district heating system annual heat generation, the total prorated water consumption from the district heating system is 500,000 gallons x 1% = 5,000 gallons. This must represent at least 10% of the total building water consumption, meaning that the total building consumption must be less than or equal to 50,000 gallons.

Further Explanation

Required Documentation

Documentation	Option 1		Option 2	Option 3
	1 point	2 points		
Potable water analysis results	x	X		
Potable water analysis narrative	x	X		
Cycles of concentration calculations	x	X		
Recycled Nonpotable water calculations		X		
Water treatment calculations		X		
Nonpotable water analysis (if using 100% nonpotable water)		X		
Documentation showing that project Baseline system is designated as systems 7, 8, 11, 12, or 13 under ASHRAE 90.1-2016 Appendix G Table G3.1.3			x	
Calculations demonstrating percent reduction in cooling tower water usage for systems where the proposed design uses the latent heat of evaporative cooling of water			x	
Site or mechanical systems plan, energy model or other showing project design			x	x
Water subsystem monthly demand calculations				x
Recycled alternative water source quantity calculations and plumbing drawings/schematics of the alternative water system. For municipally supplied alternative water, provide documentation that the municipality has agreed to supply the volume of recycled alternative water claimed by the project				x
Manufacturer information for water subsystem				x

Exemplary Performance

- Option 2. Demonstrate a 100% reduction in annual water use compared to a water-cooled chiller system.
- Option 3. Use minimum 40% recycled alternative water to meet process water demand (all projects except Core and Shell). Core and Shell: use minimum 50% recycled alternative water to meet process water demand.

Referenced Standards

- ASHRAE Standard 90.1-2016

Connection to Ongoing Performance

- LEED O+M WE credit Water Performance: Designing building cooling systems and other water subsystems to minimize potable water and reuse alternative water sources can significantly reduce the project's water footprint over the building life cycle, which may help improve a project's water performance score. Additionally, treating and maintaining the quality of makeup water used to meet process water demands can preserve the performance and efficiency of water using subsystems, reducing the frequency of replacement and repairs.